REVISION	REV	DATE	REMARKS	PREPARED BY	CHKD BY
	0	24.01.12	FOR APPROVAL	JASPREET	S.BHATIA

CUSTOMER:	MONNET POWER COMPANY LIMITED
CONSULTANT:	**DEVELOPMENT CONSULTANT PVT. LTD. KOLKATA** **CONSULTING ENGINEERS**
PROJECT:	2X525 MW THERMAL POWER STATION MALIBRAHMANI, ANGUL, ORISSA
PRIME CONTRACTOR:	BHARAT HEAVY ELECTRICALS LTD POWER SECTOR PROJECT ENGINEERING MANAGEMENT NOIDA
COLLABORATOR:	FUJIAN LONGKING, LTD.
PLANT SUPPLIER:	MELCO INDIA PVT. LTD. PLOT NO.–4, SECTOR–27A FARIDABAD–121 003

	NAME	SIGN	DATE
DRN	VIDYA		
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TITLE: FAT Procedure of PLC Panel

DEPT. | SCALE: 1:1 | SIGN

DOCUMENT NO. PE-VO-342-160-A406

SHEET 24 | REV. 0

FAT

CLIENT: MELCO INDIA PVT LTD

PROJECT : 2X525 MW THERMAL POWER PLANT, MALIBRAHMANI, ANGUL, ORISSA

PLC & SCADA BASED CONTROL SYSTEM

FACTORY ACCEPTANCE TEST

PROCEDURE

REVISION NO.- REV0 MIPL-11-01-E-06 PE-VO-342-160-A406	TITLE - FACTORY ACCEPTANCE TESTS PROCEDURE	PREPARED BY – MAX ENGINEERING & AUTOMATION P. LTD	DATE
		REMARK / SIGNATURE	
END USER	Monnet Power Company LTd, Malibrahmani , Angul , Orissa Thermal Power Plant		
CONSULTANT	DCPL, Kolkata		
CONTRACTOR	M/S Bharat Heavy Electricals Limited		
CLIENT	M/S MELCO INDIA PVT LTD		
PROJECT NAME : Mill Reject Handling System for 2X525 MW TPP			

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SECTION 0 : REVIEW OF DOCUMENTS AND TOOLS

0.1 REVIEW OF 'CONSTRUCTION' DOCUMENTS

OUTLINE

The purpose of this test is to review the documents on the basis of which the inspection is conducted..

PROCEDURE

Check that the 'Construction' drawings have been established on the basis of the **approved drawings** transmitted at the time of inspection call.
Check that the modifications carried by the 'Construction' drawings are marked in blue pen and dated and signed by the engineer-in- charge.
Check that the modifications carried in the drawings do not affect the design.

CRITERIA
The 'construction' documents show the same revision number that the documents submitted at the inspection call.

COMMENTS

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SECTION 1 : PLC I/O HARDWARE FUNCTIONING

1.1 PANEL EARTHING

OUTLINE

The purpose of this test is to insure the safety of the personnel while handling the panel, and safety of the installed equipment from any electrical hazards/ short – circuits.

PROCEDURE

Check that the panel earth point is at the building earth.

Check that the 0V inside the panel is connected to the earth bar.
Check that the panel chassis is connected to the earth bar (multi-meter)

CRITERIA

The multi-meter shows continuity between panel earth bar and 0V
The multi-meter shows continuity between panel earth bar and panel chassis.

COMMENTS

SECTION 1 : PLC I/O HARDWARE FUNCTIONING

1.2– HARDWARE CHECK

1.2.1- PLC HARDWARE

OUTLINE All hardware components for which test certificates exist are subjected to complete testing before being staged for inspection.

OBJECTIVE The purpose of this test is to verify test certificates for the following items exist :

1. Input/ Output modules
2. I/O Rack
3. PLC processor system
4. Communication modules

REFERENCE

1. PLC system configuration as per **approved drawing.**
2. Bill of material as per **approved drawing**

PROCEDURE

1. Check that the tests certificates are available for above items

CRITERIA

1. Certificate of conformity with serial nos. for PLC modules are available.
2. Certificate of conformity of operation of PLC modules

COMMENTS

SECTION 1 : PLC HARDWARE I/O FUNCTIONING

1.2– HARDWARE CHECK

1.2.2- ELECTRICAL PROTECTIONS

OUTLINE
All protection devices show a rated capacity in conformity with the design.

OBJECTIVE
The purpose of this test is to verify proper protection as per design.

1. MCBs
2. Fuses

REFERENCE
1. Panel internal wiring diagram
2. Product data sheet

PROCEDURE
1. Compare the rated capacity of each protection device in the product data sheet and the rating indicated in the panel internal wiring diagram **as per approved drawings**

CRITERIA
1. Exact match with the rating indicated in the panel internal wiring diagram

COMMENTS

SECTION 1 : PLC I/O HARDWARE FUNCTIONING

1.3– PLC I/O ARCHITECTURE

OUTLINE
To Verify System Architecture.

OBJECTIVE
The purpose of this test is to verify that :

1. The System architecture is conforming to the approved one.

REFERENCE

1. 1. Approved PLC System Architecture **drawing no**

PROCEDURE

2. Visually check that processors & I/O's are connected on EHTERNET Bus as per approved system architecture
3. Visually check the availability of the Ethernet port for connection to Ethernet network, being used for Programming & SCADA..

CRITERIA
The quantity, location & physical connectivity of these items are in accordance with the approved system architecture.

COMMENTS

SECTION 1 : PLC I/O HARDWARE FUNCTIONING

1.4 – I/O LOADING SPECIFICATION

OUTLINE
All I/O modules and communication modules are assigned during detailed engineering, and quantity/position of each item installed in the PLC panel are of utmost importance.

OBJECTIVE
The purpose of this test is to verify quantity, mounting & location of :

1. Input/ Output modules
2. I/O Rack
3. Rack Power supply
4. PLC processor system
5. Communication modules

REFERENCE

1. Approved PLC system configuration **drawing no**.
2. PLC panel general arrangement **drawing no.**
3. Internal wiring diagram and IO interconnection drawings.

PROCEDURE

1. Check the rack power supply input voltage rating,
2. Visually check that each I/O base is correct in quantity and installed in the correct position
3. Visually check that all labeling is in order

CRITERIA
The quantity and location of the I/O bases are in accordance with the system configuration.
COMMENTS

SECTION 1 : PLC I/O HARDWARE FUNCTIONING

1.5– PLC POWER SUPPLY CHECK

OUTLINE
The power supplies incoming to the panel and generated inside the panel are conforming to the design.

OBJECTIVE
The purpose of this test is to ensure that the control panel is powered in normal conditions and the control panel delivers interrogation voltage in conformity with the design.

REFERENCE
1. Internal wiring diagram
2. Power supply data sheet

PROCEDURE
1. Measure the voltage of incoming power supplies.
2. Measure the voltage of power supplies generated inside the panel as per approved data sheet and approved wiring diagram no

CRITERIA
1. The incoming voltage is matching the value indicated in the internal wiring diagram with a tolerance of +/- 10%

FAT

COMMENTS

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END USER	Monnet Power Company LTd, Malibrahmani , Angul , Orissa Thermal Power Plant		
CONSULTANT	DCPL, Kolkata		
CONTRACTOR	M/S Bharat Heavy Electricals Limited		
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PROJECT NAME : Mill Reject Handling System for 2X525 MW TPP			

SECTION 1 : PLC I/O HARDWARE FUNCTIONING

1.6– PLC START-UP

OUTLINE
PLC goes to normal operation on powering up, provided that all application software are stored in respective PLC's.

OBJECTIVE
The purpose of this test is to ensure that PLC automatically boots up from Power off to operation .

PROCEDURE

1. Power on the entire system
2. Power off the PLC
3. Power on the PLC again

CRITERIA
The PLC goes to Run mode

COMMENTS

SECTION 1 : PLC I/O HARDWARE FUNCTIONING

1.7 I/O FUNCTION CHECK

1.7.1 DIGITAL INPUTS

OUTLINE
All Digital Inputs should be terminated at the proper terminal only.

OBJECTIVE
To check Input wiring.

REFERENCE
Approved Panel Internal wiring diagram

PROCEDURE

1. Connect jumper to digital input field termination block as per wiring drawings to toggle field input.
2. Observe LED indication on PLC Input modules.
3. Observe corresponding status in the PLC memory on programming terminal.
4. Observe corresponding control on scada screen.

CRITERIA

1. Digital Input Indication as per internal wiring diagram.

COMMENTS

SECTION 1 : PLC HARDWARE I/O FUNCTIONING

1.7 I/O OPERATION CHECK

1.7.2 DIGITAL OUTPUTS

OUTLINE
All Digital Outputs should be terminated at the proper terminal only.

OBJECTIVE
To check Digital Output wiring.

REFERENCE
Approved Panel Internal Wiring Diagram

PROCEDURE
1. Force the Output with the help of programming software
2. Observe LED Indication in the corresponding digital output termination module
3. Check Relay Operation and check the contacts assigned for particular output.
4. Observe corresponding control on scada screen..

CRITERIA
Digital Output Indication as per Internal Wiring Diagram.

COMMENTS

SECTION 1 : PLC HARDWARE I/O FUNCTIONING

1.7 I/O OPERATION CHECK

1.7.3 ANALOG INPUTS

OUTLINE
All Analog Inputs should be terminated at the proper terminal only.

OBJECTIVE
To check Analog Input wiring.

REFERENCE

1. Approved Panel Internal Wiring Diagram
2. PLC analogue input module data sheet

PROCEDURE

1. Measure the current provided by the source with a multimeter
2. Observe corresponding value on scada screen.

CRITERIA

1. Value read in the PLC is in the correct I/O memory location

COMMENTS

SECTION 1 : PLC HARDWARE I/O FUNCTIONING

1.7 I/O OPERATION CHECK

1.7.4 ANALOG OUTPUTS

OUTLINE
All Analog Outputs should be terminated at the proper terminal only.

OBJECTIVE
To check Analog Output wiring.

REFERENCE

1. Approved Panel Internal Wiring Diagram
2. PLC analog output module data sheet

PROCEDURE

1. Force the Output memory location with the help of programming software.
2. Observe corresponding value on scada screen

CRITERIA

1. Value read is correct in terms of the specified terminals corresponding to the forced value I/O memory location

COMMENTS

SECTION 2 : PLC COMMUNICATION HARDWARE FUNCTIONING

2.1 – PLC PROGRAMMING PORT

OUTLINE
Operation of the programming terminal through the PLC programming port.

OBJECTIVE
The purpose of this test is to verify that an operator can take control of the system of the PLC via a programming terminal. The objective is to verify :

1. Correct operation of the PLC programming port
2. Correct operation of the system of the CPU

.

PROCEDURE

1. Connect the programming terminal to the programming port of the PLC
2. Force I/O from the programming terminal.
3. Add logic while processor is running
4. Delete logic at the end of the test

CRITERIA

1. Check that the corresponding I/O is forced.
2. Check logic is performing according to the change.

COMMENTS

SECTION 2 : PLC COMMUNICATION HARDWARE FUNCTIONING

2.2 – PLC COMMUNICATION PORTS

OUTLINE
Connection of the programming terminal to the PLC through all available connection points : Ethernet .

OBJECTIVE
The purpose of this test is to verify that the Ethernet communication ports is working.

PROCEDURE

1. Connect the programming terminal to all available connection points and laptop.
2. Check that the programming terminal connects to the PLC CPU.

CRITERIA

1. Check that the programming terminal Runs/Stops the PLC from each connection point

COMMENTS

SECTION 3 : RELAYS AND RELAY BOARD TEST

3 .1 FUNCTIONAL TEST OF RELAYS

OUTLINE

All the Relays are functioning and the terminations as per requirements.

OBJECTIVE

The purpose of this test is to test the functionality of Relays and it's change over contact.

PROCEDURE

1. Apply 24 V DC supply to coil of relay by jumper either by forcing the PLC outputs. .
2. Check the change of state by multi meter across relay contacts.

CRITERIA

1. NO contact of relay will become NC and vice versa. Contact change over shall be tested by continuity tester/ multi-meter.

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SECTION 4 : PLC SYSTEM CHECK

4.1 – I/O SIMULATION CHECK

OUTLINE The following functions are tested :

1. Forcing of I/Os

OBJECTIVE

The purpose of this test is to verify that the console takes control of the PLC program.

PROCEDURE

1. Force I/O from the programming terminal.

CRITERIA

Check that the corresponding I/O is forced.

4.2 – CPU/ IO NETWORK REDUNDANCY CHECK

OUTLINE The following functions are tested :

1. CPU REDUNDANCY
2. IO Network Redundancy

OBJECTIVE The purpose of this test is to verify that the CPUs and network are redundant

REFERENCE

1. Approved Panel System Configuration **drawing** .

PROCEDURE

1. To switch off the Primary PLC and see whether the network switches to secondary PLC

4.3 – MMI (SCADA) CHECK

OUTLINE The following functions are tested :

To check SCADA screens
To check security feature
To check Alarm history
To check trending

OBJECTIVE The purpose of these tests are to verify that the Screens, Security feature, data archiving, alarms, trending and report generation are made as per customer requirement.

REFERENCE

1. Approved P&ID Diagram.

4.4 – TRIP/INTERLOCK LOGIC CHECK

OUTLINE

All trip functions are working properly

OBJECTIVE The purpose of this test is to verify that the console takes control of the PLC program.

REFERENCE

1. Approved control philosophy.

PROCEDURE

1. Force Fault Condition

CRITERIA

Check whether all the trips indications are working or not.

4.5 – GA/DIMENSION/BOM/SPARE CHECK

OUTLINE The following functions are tested :

3. GA
4. BOM
5. DIMENSIONS
6. SPARES

OBJECTIVE The purpose of this test is to verify that the all the above mentioned are according to the drawing

REFERENCE

1. Approved GA drawing no. **2107-301-06**

4.6 – ONLINE SYSTEM CONFIG CHECK

OUTLINE The following functions are tested :

1. check system config

OBJECTIVE The purpose of this test is to verify that the System Config in online state is correct

PROCEDURE

1. check the communication at all available ports.
Check logic is performing according to the change.

COMMENTS

SECTION 5 : TESTING OF SOFTWARE

5.1 TESTING OF PROGRAM LOGIC

OUTLINE

PLC Program developed as per requirement of system process in ladder or FBD language.

OBJECTIVE

The purpose of this test is to verify the digital interlocks as per to the Operational Logic.

PROCEDURE

Force the inputs and the required interlocks are checked.

REFERENCE

1. Approved control philosophy.

COMMENTS

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